THE SCHEME THEORETIC IMAGE OF J = (T, S) FOR A QUASI-COMPACT GROUPOID SCHEME OVER A FIELD CARRIES THE RESTRICTED GROUPOID STRUCTURE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition of the scheme theoretic image of a morphism (morphisms chapter, Definition morphisms-definition-scheme-theoretic-image, canonical tag 01R7; kind definition). cited at statement lines 1975-1977. Defines Z as the scheme theoretic image of j = (t, s) : R to U times_S U, the subscheme carrying the restricted groupoid structure.
- The quasi-compact map groupoids-on-field-image lemma, of which the groupoid case is a special instance (more-groupoids chapter, Lemma lemma-quasi-compact-map-groupoids-on-field-image, canonical tag 04QA). cited at proof lines 1987-1989. One route to the result: the ambient groupoid (U, U times_S U, pr_1, pr_0, pr_02) makes the statement a special case.
- Scheme theoretic image of a product of morphisms over a field equals the product of the scheme theoretic images (varieties chapter, Lemma varieties-lemma-scheme-theoretic-image-product-map, canonical tag 04Q1). cited at proof lines 2013-2016. Identifies the scheme theoretic image of j_s times _tj with Z_s times_k _tZ, which is the key input for the containment of Z_s times_k _tZ in Z.
- The perfect-field version of the image groupoid lemma, whose proof handles sigma-invariance, the diagonal factorization and the remaining groupoid axioms (more-groupoids chapter, Lemma lemma-groupoid-on-perfect-field-image, canonical tag 04QC). cited at proof lines 2018-2022. Template for the concluding steps: sigma(Z) inside Z, the diagonal of U over S factoring through Z, and the remaining axioms, none of which uses quasi-compactness.

Source reference. Original definition 01R7: sources/stacks-chow/tags.txt, source lines 2292–2292. Cited at statement lines 1975-1977: defines Z as the scheme theoretic image of j = (t, s).

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Source reference. Original lemma 04QA: sources/stacks-chow/tags.txt, source lines 6143–6143. Cited at proof lines 1987-1989: the ambient groupoid (U, U times_S U, pr_1, pr_0, pr_02) makes the statement a special case of this lemma.

Source reference. Original lemma 04Q1: sources/stacks-chow/tags.txt, source lines 6134–6134. Cited at proof lines 2013-2016: the scheme theoretic image of j_s times _tj is Z_s times_k _tZ, the key input for the containment in Z.

Source reference. Original lemma 04QC: sources/stacks-chow/tags.txt, source lines 6145–6145. Cited at proof lines 2018-2022: its written proof is the template for sigma(Z) inside Z, the diagonal factorization, and the remaining groupoid axioms.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume U is the spectrum of a field and assume R is quasi-compact (equivalently s, t are quasi-compact). Let $Z \subset U \times_S U$ be the scheme theoretic image (see Morphisms, Definition 01R7) of $j = (t, s) : R \to U \times_S U$. Then*

$$(U, Z, pr_0|_Z, pr_1|_Z, pr_{02}|_{Z \times_{pr_1, U, pr_0} Z})$$

is a groupoid scheme over S.

Proof. As $(U, U \times_S U, \mathrm{pr}_1, \mathrm{pr}_0, \mathrm{pr}_{02})$ is a groupoid scheme over S this is a special case of Lemma 04QA. But we can also prove it directly as follows.

The main difficulty is to show that $\mathrm{pr}_{02}|_{Z \times_{\mathrm{pr}_1, U, \mathrm{pr}_0} Z}$ maps into Z. Write $U = \mathrm{Spec}(k)$. Denote R_s (resp. Z_s, resp. U_s^2) the scheme R (resp. Z, resp. $U \times_S U$) viewed as a scheme over k via s (resp. $\mathrm{pr}_1|_Z$, resp. pr_1). Similarly, denote ${}_tR$ (resp. ${}_tZ$, resp. ${}_tU^2$) the scheme R (resp. Z, resp. $U \times_S U$) viewed as a scheme over k via t (resp. $\mathrm{pr}_0|_Z$, resp. pr_0). The morphism j induces morphisms of schemes $j_s : R_s \to U_s^2$ and ${}_tj : {}_tR \to {}_tU^2$ over k. Consider the commutative diagram

$$\begin{array}{ccc} R_s \times_{k\ {}_tR} & \xrightarrow{c} & R \\ \downarrow{\scriptstyle j_s \times_t j} & & \downarrow{\scriptstyle j} \\ U_s^2 \times_{k\ {}_tU^2} & \longrightarrow & U \times_S U \end{array}$$

By Varieties, Lemma 04Q1 we see that the scheme theoretic image of $j_s \times {}_tj$ is $Z_s \times_{k\ {}_tZ}$. By the commutativity of the diagram we conclude that $Z_s \times_{k\ {}_tZ}$ maps into Z by the bottom horizontal arrow. As in the proof of Lemma 04QC it is also true that $\sigma(Z) \subset Z$ and that $\Delta_{U/S}$ factors through Z. Hence we conclude as in the proof of that lemma. □